



BTP
DEPARTMENT OF MECHANICAL ENGINEERING

Investigation of Ti-Nb-Zr-Ta bio-metallic alloy

Under the guidance of:
Dr. Dan Sathiaraj (Associate Professor)

Presented By:

Name	Roll Number
M. Naveen	230003042
R. Manthru Naik	230003055

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Background: Titanium Alloys for Biomedical Implants

❑ Need for Biomedical Implant Materials

- Biomedical implants require **high strength, corrosion resistance, biocompatibility and long-term stability**. [\[1,2,3\]](#)
- **Titanium alloys** are widely used because of their **good mechanical properties, corrosion resistance and biocompatibility**. [\[1,2,3\]](#)
- **Commercially pure Ti may have insufficient strength** for some load-bearing applications, motivating the development of stronger Ti alloys. [\[1\]](#)

❑ Limitations of Conventional Ti Alloys

- **Ti-6Al-4V** provides high strength, but concerns regarding conventional alloying elements have encouraged the development of alternative Ti alloys. [\[1,3\]](#) A key limitation is the **large elastic-modulus mismatch between Ti alloys and bone**. [\[3\]](#)
- A major limitation is the **large elastic-modulus mismatch between Ti alloys and bone**, which can contribute to **stress shielding**. [\[1\]](#)
- **Cortical bone ≈ 30 GPa**, whereas **Ti-6Al-4V ≈ 112 GPa**, highlighting the need for lower-modulus implant materials. [\[1\]](#)

Background: Development of β -Titanium Alloys

❑ β -Ti Alloys as a Promising Solution

- β -Ti alloys are stabilized using β -stabilizing elements such as Nb and Ta. [\[1,3\]](#)
- Compared with conventional α and $\alpha+\beta$ Ti alloys, β -Ti alloys offer: [\[1\]](#)
 - Lower elastic modulus
 - Good corrosion resistance
 - Good biocompatibility
- The β phase has a BCC crystal structure and its stability strongly influences the **elastic modulus and mechanical properties** of the alloy. [\[1,2\]](#)
- Therefore, **alloy composition and processing conditions** are important for controlling the **phase constitution, microstructure and properties** of β -Ti alloys. [\[1,3\]](#)

❑ Why Ti–Nb–Ta–Zr?

- Nb, Ta and Zr are attractive alloying elements for biomedical Ti alloys because of their **biocompatibility and favourable corrosion behaviour**. [\[1,3\]](#)
- Ti–Nb–Ta–Zr (TNTZ) alloys are investigated as β -Ti alloys because they combine **low elastic modulus, corrosion resistance and biomedical compatibility**. [\[1,3,4\]](#)
- Ti–30Nb–10Ta–5Zr has been reported to exhibit a **predominantly/pure β -Ti phase** and good biomedical potential. [\[2\]](#)

Literature Review

No.	Alloy / Focus	Key Observation / Findings
1	β -Ti alloys / Biomedical applications [1]	✓ β-Ti alloys offer low elastic modulus and good biocompatibility. ✓ Alloy design, processing, microstructure, mechanical properties, corrosion behaviour and biocompatibility influence their biomedical performance
2	Ti-30Nb-10Ta-5Zr [2]	✓ Alloy exhibited a pure β-Ti crystalline phase with suitable mechanical properties. ✓ Good biocompatibility was demonstrated through cell attachment, spreading and implantation studies.
3	Ti-Nb-Zr-Ta / Corrosion & Biocompatibility [3]	✓ As-cast TNZT showed a predominant β-Ti phase with relatively low elastic modulus. ✓ The alloy showed low ion release, good cell adhesion and no cytotoxicity, indicating good biocompatibility.
4	Ti-35Nb-7Zr-5Ta [4]	✓ EBM-produced alloy showed predominantly bcc β-Ti phase. ✓ PEB treatment improved surface topography, while bulk mechanical properties and corrosion resistance showed no significant change.
5	Ti-35Nb-7Zr-5Ta / Corrosion behaviour [5]	✓ HIP processing produced a highly dense β-Ti alloy with improved β-phase homogeneity. ✓ Higher cooling rate during HIP improved corrosion resistance and reduced metal-ion release.
6	TiNbZrTa / Bio-corrosion & biocompatibility [6]	✓ TiNbZrTa exhibited a single BCC structure with a wider and more stable passivation region than TiNbZrTaMo. ✓ TiNbZrTa showed good biocompatibility, with lower cytotoxicity than the Mo-containing alloy.

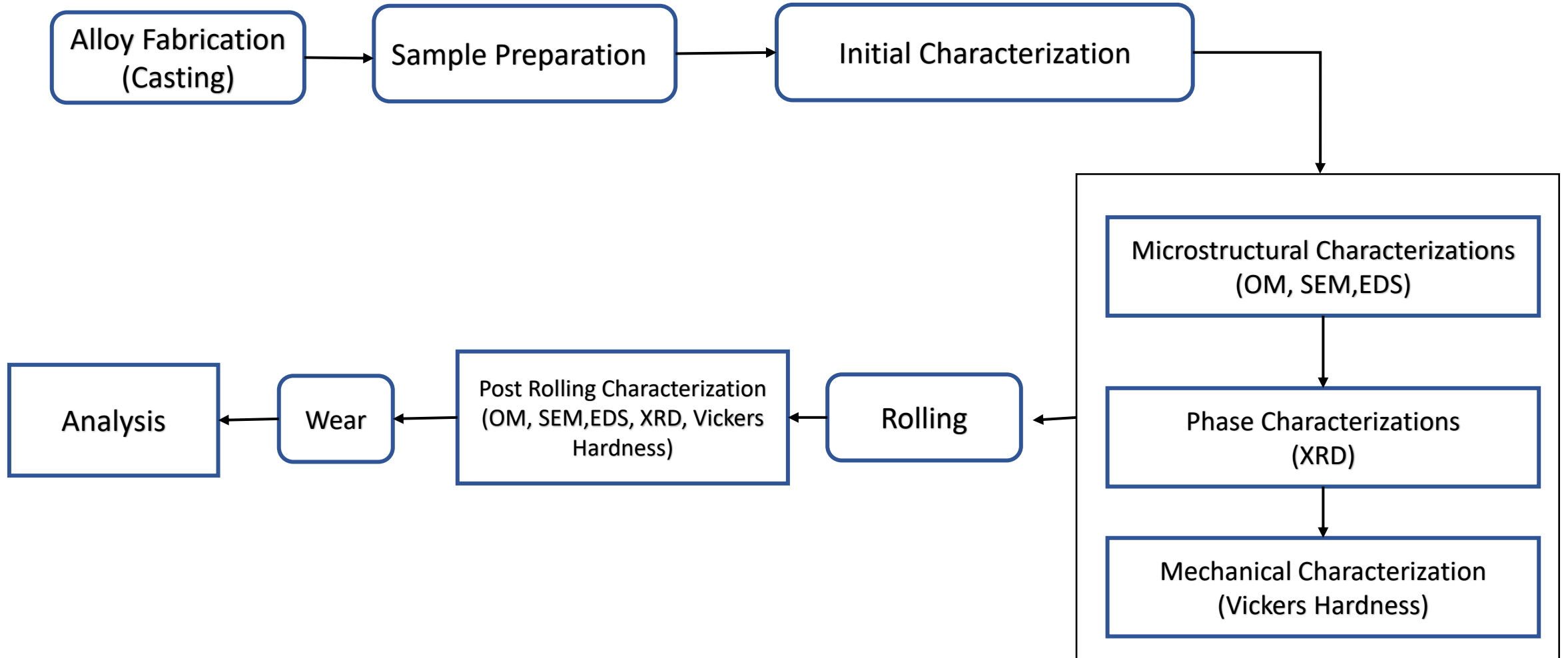
Objectives

- ❑ **To investigate the microstructural, compositional and mechanical properties** of the Ti-based bio-metallic alloy using optical microscopy, SEM, EDX, XRD and Vickers hardness testing.

- ❑ **To investigate the wear and corrosion behaviour** of the alloy and establish their **relationship with the material's microstructure and properties.**

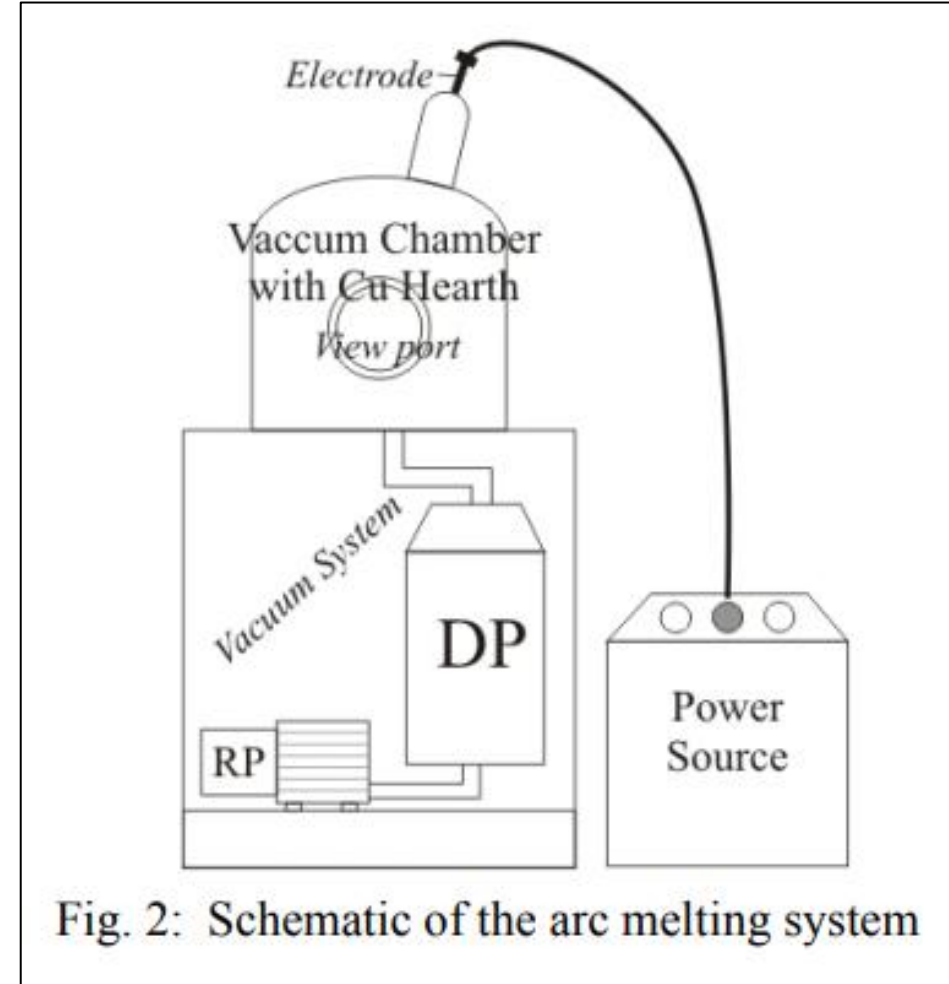
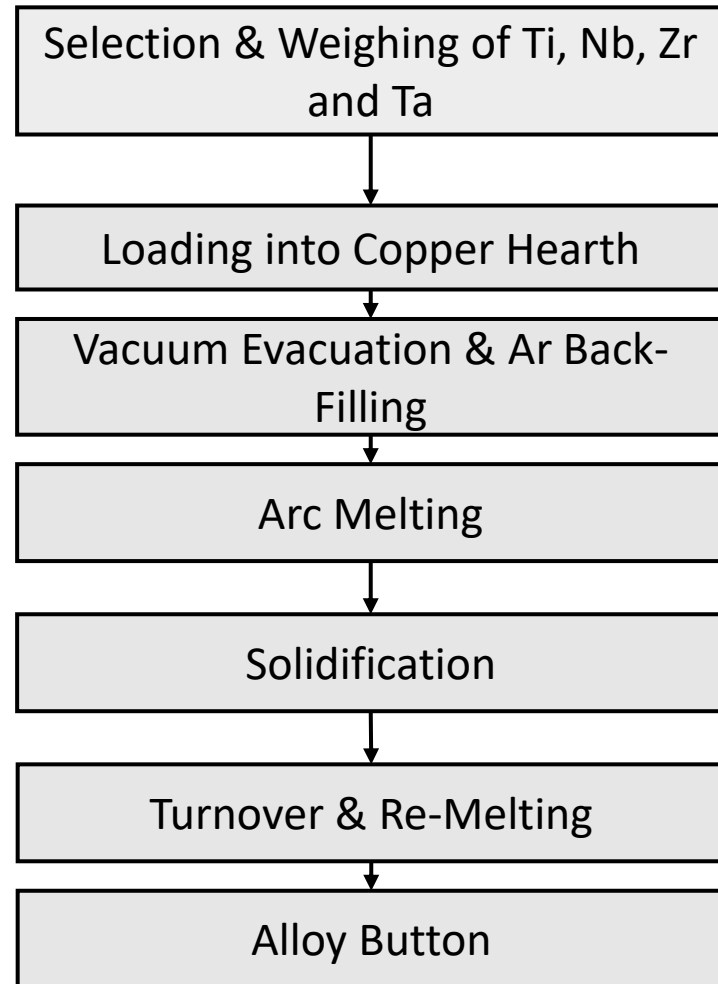
- ❑ **Alloy Compositions :**
 1. **Ti–30Nb–5Zr–10Ta**
 2. **Ti–35Nb–5Zr–5Ta**

Methodology



Alloy Preparation – Vacuum Arc Melting

❑ **Purpose:** Formation of the alloy with improved compositional homogeneity and minimized oxidation.



Experimental Work

1. Sample Preparation

❑ Grinding & Polishing

- Removes surface irregularities and scratches to obtain a smooth, scratch-free surface suitable for microstructural examination.



❑ Etching

- Reveals microstructural features by selectively attacking the polished surface.
- **Etching Time:** 30 sec

Component	Composition
Distilled Water (H ₂ O)	90%
Nitric Acid (HNO ₃)	7%
Hydrofluoric Acid (HF)	3%

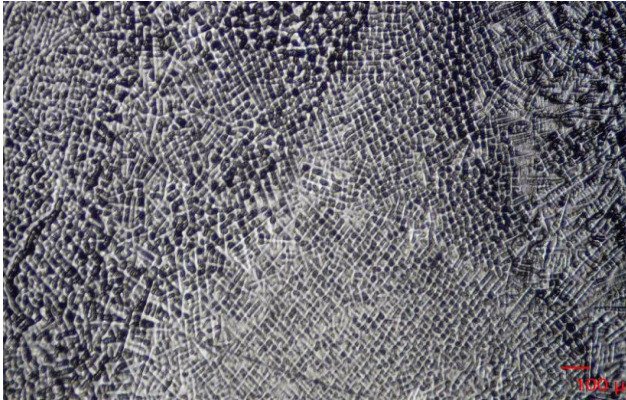


2. Optical Microstructure

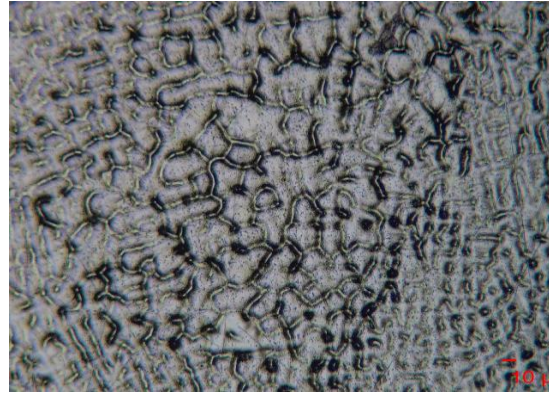
- The optical micrographs show differences in microstructural morphology between the two alloy compositions. Increasing magnification reveals finer details of the observed microstructural features.

1. Ti-30Nb-5Zr-10Ta

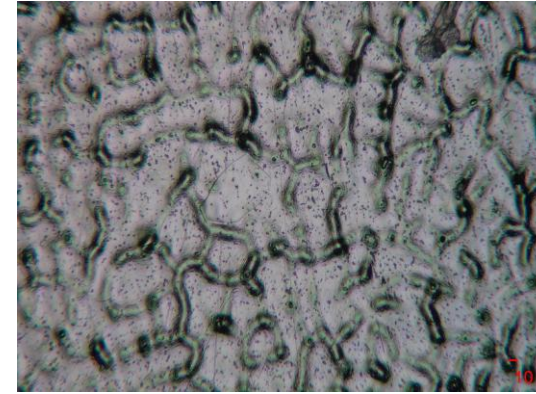
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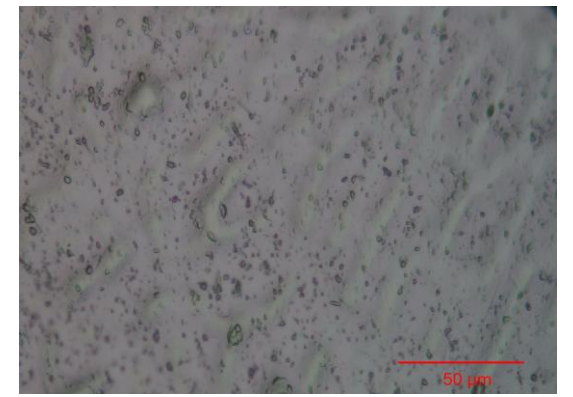
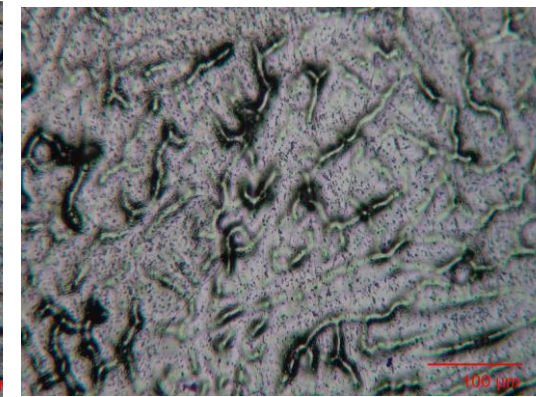
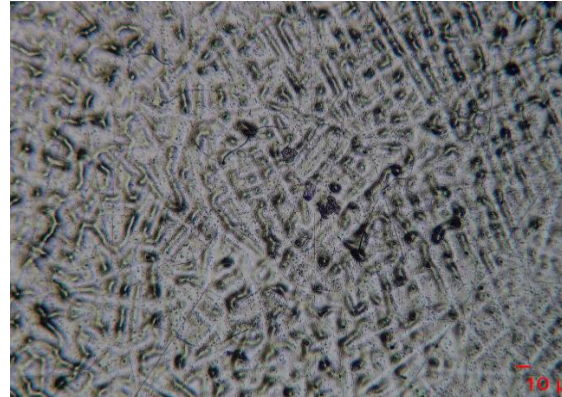
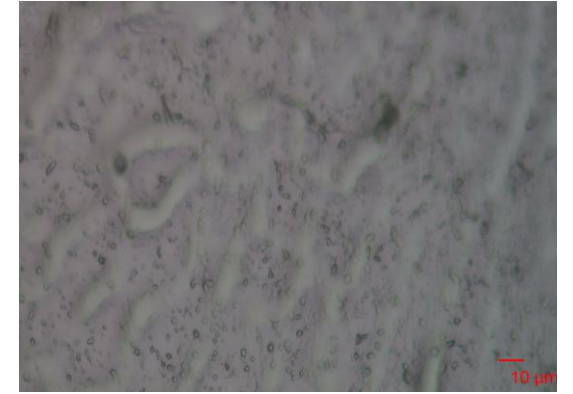
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20x

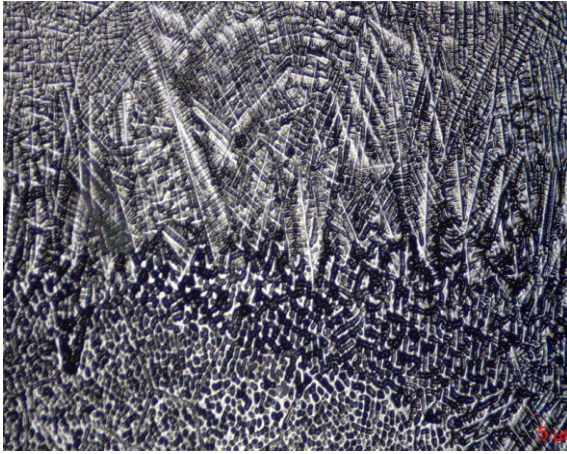


50x

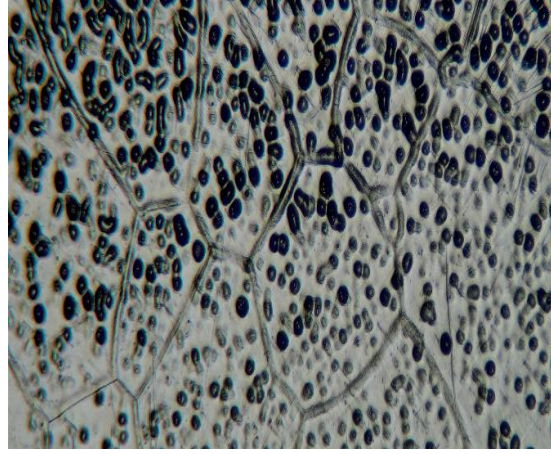


2. Ti-35Nb-5Zr-5Ta

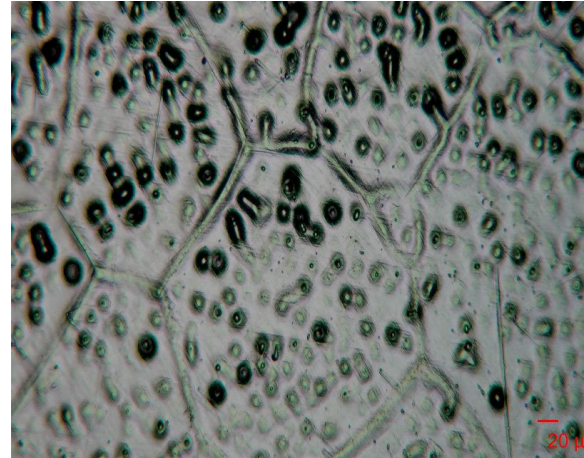
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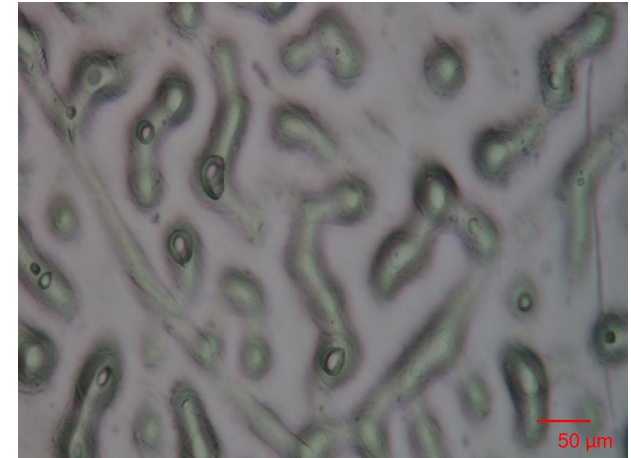
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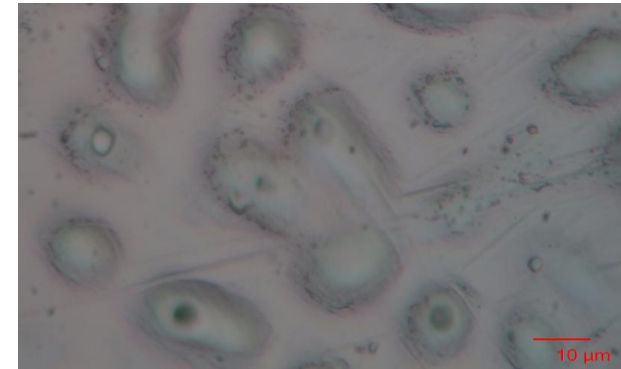
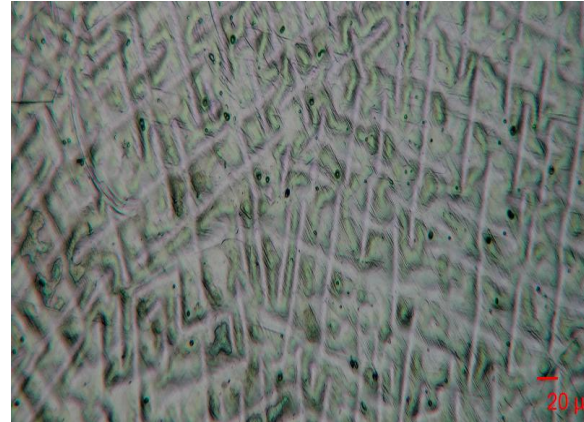
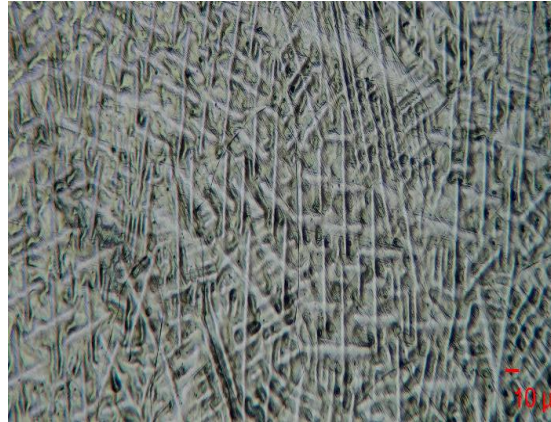
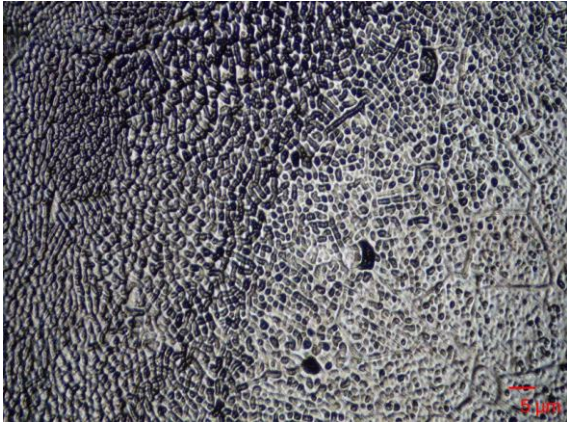
20x



50x



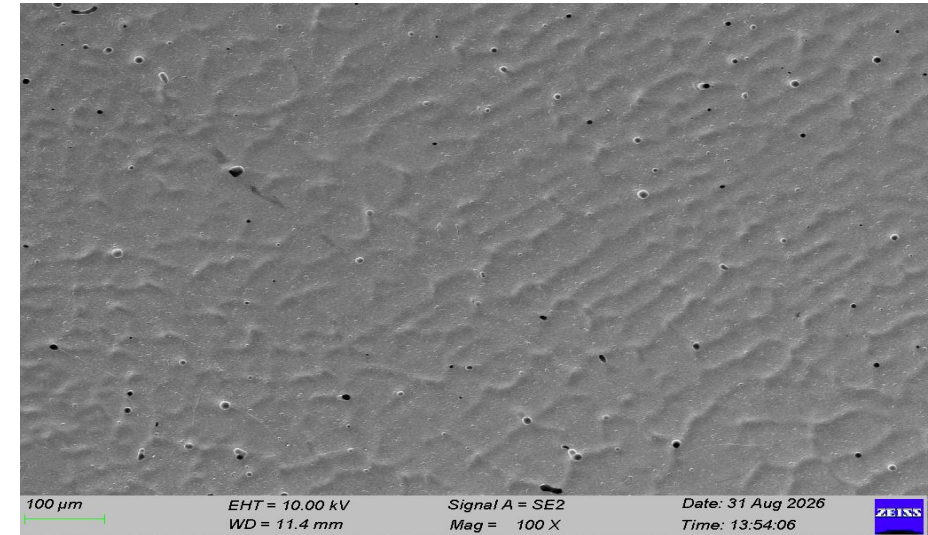
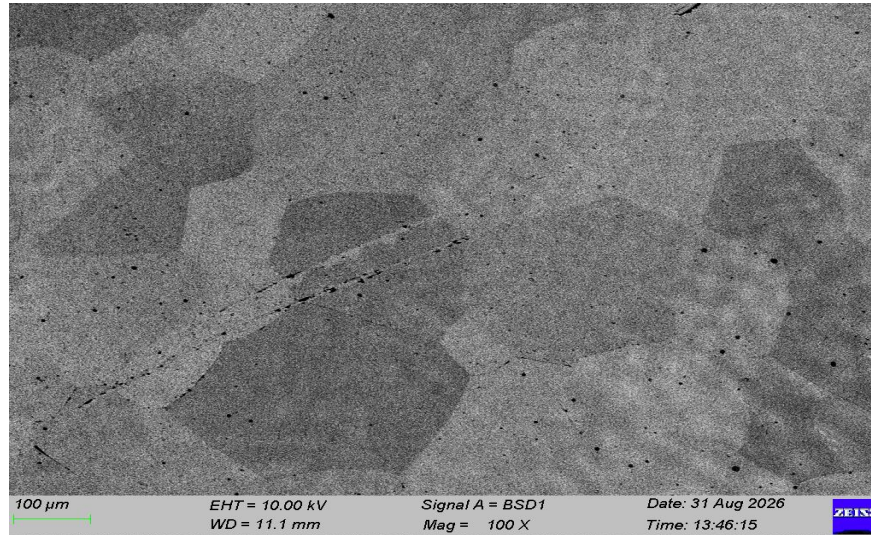
100x



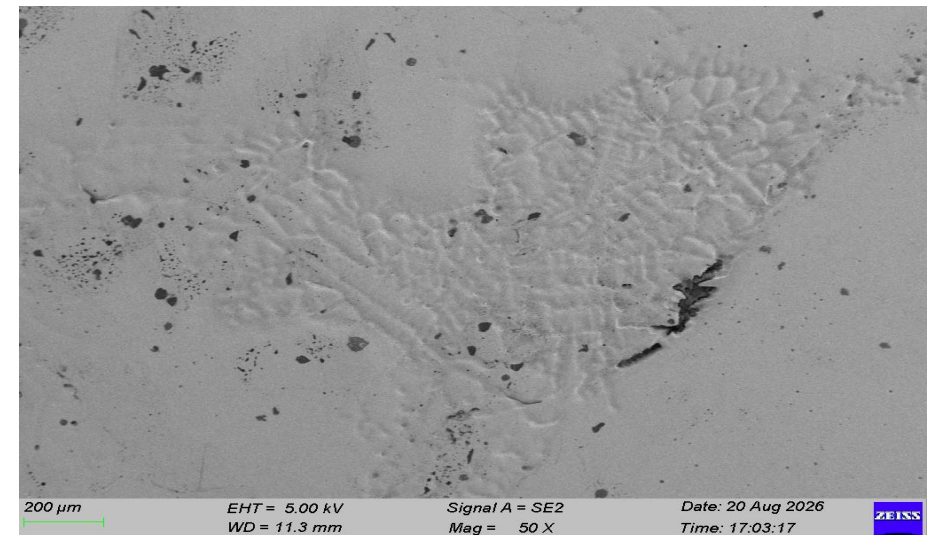
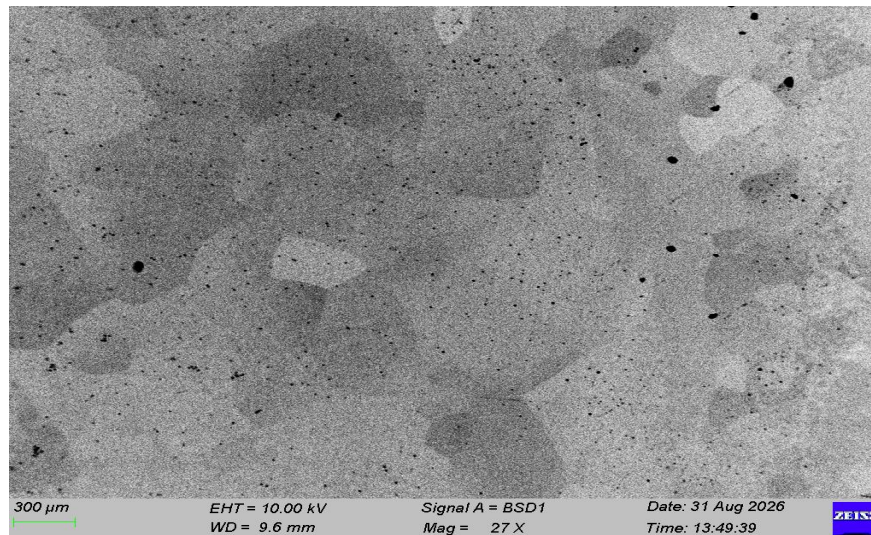
3. SEM Analysis

- ❑ Scanning Electron Microscopy (SEM) was used to examine the surface morphology and microstructural features of both Ti-based alloy samples.

Ti-30Nb-5Zr-10Ta

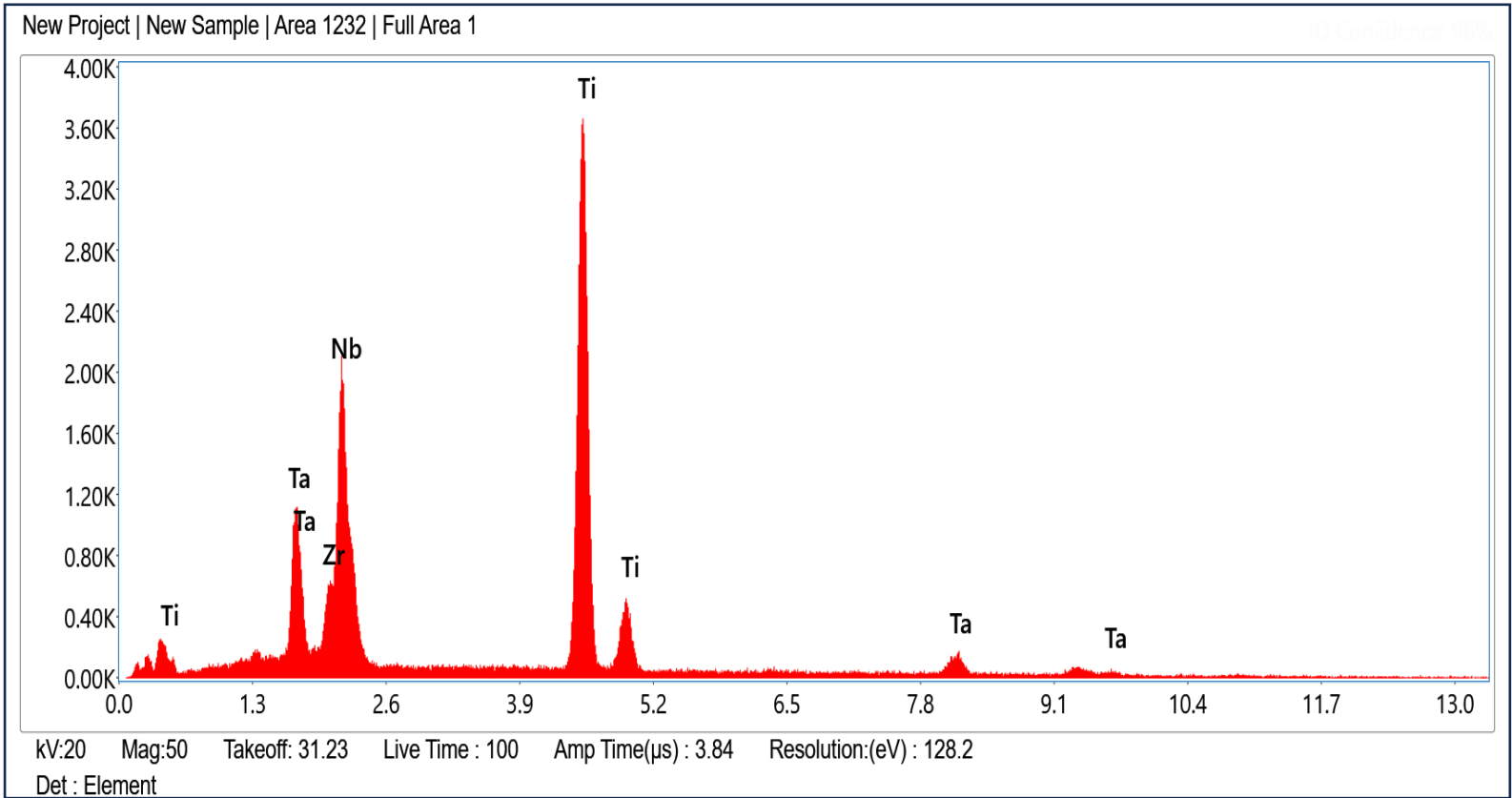


Ti-35Nb-5Zr-5Ta



4.EDS Analysis

1. Ti–30Nb–5Zr–10Ta



Element	Atomic %
Ti	58.2
Nb	28.6
Zr	6.6
Ta	6.6

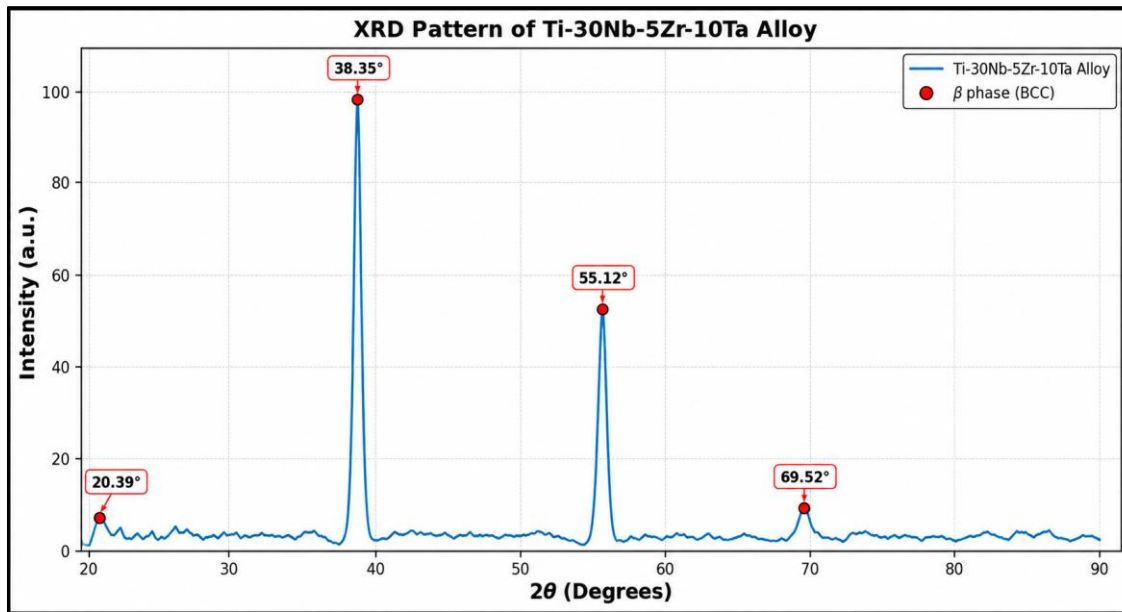
Key Observations :

- Ti, Nb, Zr and Ta were detected in the analyzed region, with Ti showing the highest atomic percentage.
- The EDX spectrum confirms the presence of the principal alloying elements Ti, Nb, Zr and Ta.

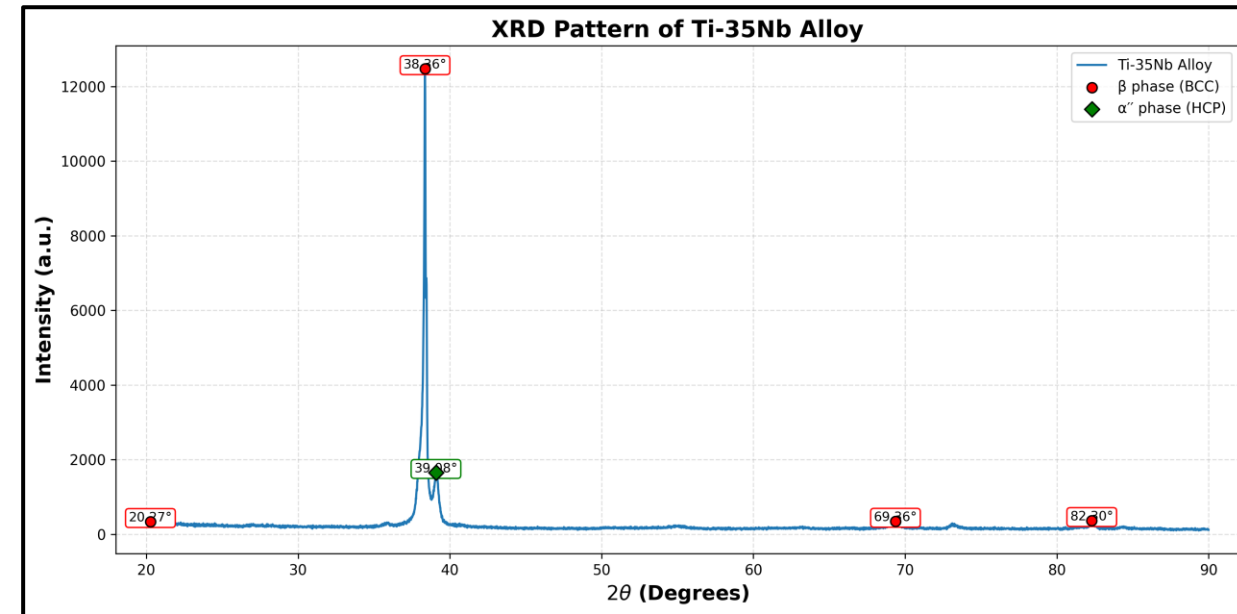
5. XRD Analysis

- ❑ **X-ray Diffraction (XRD)** is a non-destructive technique used to identify the **crystal structure, crystalline phases and structural characteristics** of a material by analyzing its diffraction pattern.

1. Ti-30Nb-5Zr-10Ta



2. Ti-35Nb-5Zr-5Ta



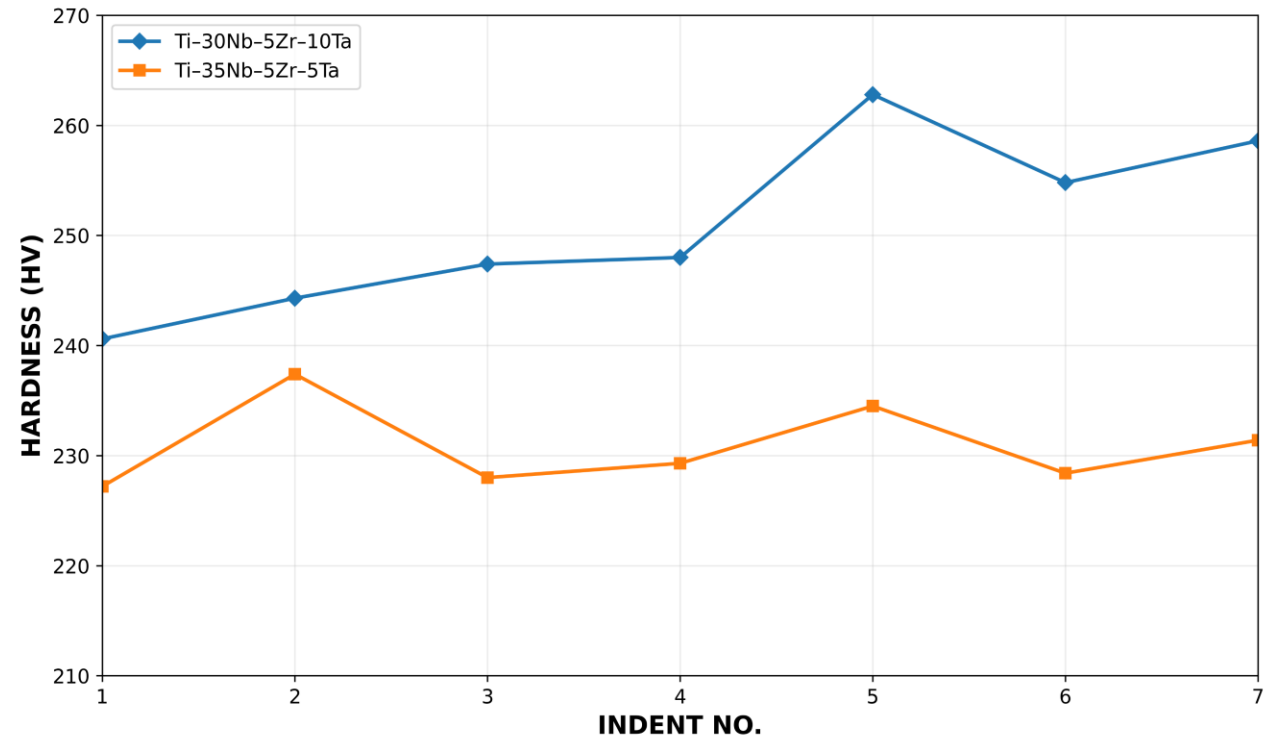
Key Observations :

- Both alloys show a predominant β-Ti (BCC) phase. The Ti-35Nb-5Zr-5Ta alloy also shows an α'' orthorhombic martensitic phase.[\[3,6\]](#)
- Similar peak positions indicate a similar β-phase structure in both samples.

6. Vickers Hardness Analysis

Indent	Ti-30Nb-5Zr-10Ta (HV)	Ti-35Nb-5Zr-5Ta (HV)
1	240.6	227.2
2	244.3	237.4
3	247.4	228.0
4	248.0	229.3
5	262.8	234.5
6	254.8	228.4
7	258.6	231.4
Avg	250.9 ± 8.0 HV	230.9 ± 3.8 HV

Avg = Average Hardness of the Sample
Load = 0.2 kgf, Dwell Time = 15 sec



$$HV = 1.854 \times \frac{F}{d^2}$$

Where; $d = \frac{d_1 + d_2}{2}$

Vickers Hardness – Key Observations

- ❑ **Ti–30Nb–5Zr–10Ta** shows **higher hardness** compared with **Ti–35Nb–5Zr–5Ta** under the same testing conditions.
- ❑ The **higher Ta content** may contribute to **solid-solution strengthening**, which can increase resistance to localized plastic deformation.
- ❑ The **higher Nb content** can influence **β -phase stability and microstructure**, which may affect the hardness of the alloy.
- ❑ The higher hardness may be associated with differences in Ta/Nb content, phase constitution, grain structure and other microstructural features.
- ❑ **Zr** also contributes to the overall alloy behaviour and can influence **phase stability and mechanical properties**, although its content is the same in both samples



Fig. Vickers hardness machine

Future Work

☐ Rolling & Annealing

- Perform **cold/warm rolling** to introduce controlled deformation and reduce material thickness.
- Carry out **annealing treatment** to modify the microstructure and relieve deformation-induced stresses.

☐ Wear Testing

- Evaluate the **wear resistance and wear rate** of both Ti-based alloy samples.
- Study the **wear mechanisms and worn-surface morphology**.

☐ Biocompatibility Evaluation

- Assess **cell viability, cell adhesion and cell proliferation**.
- Evaluate the suitability of the alloys for **biomedical implant applications**.

☐ Microstructure–Property Correlation

- Correlate **composition, microstructure and hardness** with the observed **wear and corrosion behaviour**.

☐ Comparative Analysis

- Compare **Ti–30Nb–5Zr–10Ta** and **Ti–35Nb–5Zr–5Ta** based on their mechanical, wear, corrosion and biological properties.

Conclusion

- ❑ **Microstructural characterization shows the presence of casting microstructures with coarser grains.**
- ❑ **EDS characterization** confirmed the presence of Ti, Nb, Zr and Ta.
- ❑ **XRD analysis** revealed a predominant **β -Ti (BCC) phase** with similar diffraction patterns for both compositions.
- ❑ **Ti–30Nb–5Zr–10Ta showed higher hardness** than Ti–35Nb–5Zr–5Ta, indicating the influence of alloy composition on mechanical properties.
- ❑ **Wear, corrosion and biocompatibility studies** will be carried out in the next stage to further evaluate their suitability for biomedical applications.

References

S.No	Paper
1	Chen, L.-Y., Cui, Y.-W., Zhang, L.-C., <i>Recent Development in Beta Titanium Alloys for Biomedical Applications</i> , Metals , 2020, 10, 1139.
2	Gudkov, S. V. et al., <i>Preparation, structural and microstructural characterization of Ti–30Nb–10Ta–5Zr alloy for biomedical applications</i> , Journal of Materials Research and Technology , 2020, 9(6), 16018–16028.
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4	Surmeneva, M. et al., <i>New Ti–35Nb–7Zr–5Ta Alloy Manufacturing by Electron Beam Melting for Medical Application Followed by High Current Pulsed Electron Beam Treatment</i> , Metals , 2021, 11, 1066.
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6	Cheng, J. et al., <i>Bio-corrosion behaviors and bio-compatibilities of TiNbZrTa and TiNbZrTaMo high entropy alloys</i> , Materials Science and Engineering: C , 2021.

THANK YOU